SHRUTI APARNA

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Education

Indian Institute of Technology, Delhi Delhi
B.Tech. in Engineering Physics (Specialization in Quantum Technology) 2022 – 2026
CGPA: **8.3/10.0**

Scholastic Achievements

- **Minor in Computer Science:** Pursuing a Minor in Computer Science alongside B.Tech degree at IIT Delhi. [2026]
- **Summer Undergraduate Research Award (SURA):** Awarded by Industrial R&D Unit, IIT Delhi. [2024]
- **JEE Examinations:** Secured AIR 6565 in JEE Advanced and ranked among the top 1 percentile in JEE Main. [2022]
- **Best Project Award (MCP101):** Received the Best Project Award among more than 100 submissions. [2023]
- **Certificate of Merit (CBSE):** Bestowed for being in top 0.1% students, scoring 100% marks in Physics. [2021]

Publication

Overcoming Challenges of Partial Client Participation in Federated Learning: A Comprehensive Review Jun 2025
Mrinmay Sen, Shruti Aparna, Rohit Agarwal, Chalavadi Krishna Mohan *arxiv.org/abs/2506.02887*

- A survey of partial client participation in FL with theoretical and empirical perspectives; prepared for submission to **IEEE Transactions on Knowledge and Data Engineering (TKDE)**.

Research Experience

Research Assistant | CARE, IIT Delhi Aug 2025 – Present
Machine Learning for CV-QKD Parameter Estimation, Prof. Neelkanth Kundu

- Designed a **feature-driven neural network** to estimate channel transmittance and excess noise in optical communication links.
- Achieved accurate secret key rate (SKR) prediction across both synthetic data and real experimental fiber links.
- Demonstrated improved robustness over traditional statistical estimators, enabling more reliable performance in noisy real-world sensing environments.

Research Assistant | Optics and Photonics Center (OPC), IIT Delhi Aug 2025 – Nov 2025
Computational Imaging and Phase Retrieval for Biological Microscopy, Prof. Kedar Khare

- Developed a **constrained phase-only optimisation** framework for multipixel interferometric phase retrieval.
- Exploited spatial redundancy and sparsity to surpass independent-pixel phase estimation limits.
- Validated performance on synthetic datasets and digital holographic microscopy (DHM) images of biological samples (red blood cells).

Summer Undergraduate Research Award (SURA) Scholar Apr 2024 – Jun 2024
Design and Fabrication of Diffractive Optical Element, Dept. of Physics, IIT Delhi

- Designed phase-only diffractive optical elements using iterative phase retrieval with gradient-based refinement.
- Reduced mean absolute error in target intensity reconstruction by **67%** relative to Gerchberg–Saxton designs.
- Fabricated optimized phase masks on quartz substrates using UV lithography and experimentally validated beam shaping performance.

Research Project | Dept. of CSE, IIT Delhi Aug 2024 – Nov 2024
Visual Image Reconstruction from Human Brain Activity (fMRI) – Prof. Rahul Garg

- Enhanced a state-of-the-art pipeline for reconstructing visual stimuli from human brain activity using deep generative models.
- Integrated CLIP, VQGAN, and Bayesian priors for semantic guidance, improving perceptual similarity of reconstructions.
- Quantitatively validated improvements using SSIM and correlation-based metrics over baseline methods.

Internships

Data Science Intern | InfoEdge (Naukri.com) May 2025 – Jul 2025

SFT and DPO Finetuning For JD Prefill

- Improved structured field extraction from unstructured job descriptions by **fine-tuning LLaMA-8B**, strengthening the JD Prefill pipeline used in production.
- Designed and productionized an end to end **SFT + DPO fine-tuning pipeline**, enabling scalable onboarding of new fields and systematic correction of high-error fields via preference optimization.
- Built an automated evaluation framework with field-wise precision and regression checks, achieving consistent gains on weak fields while preserving baseline performance on strong fields.

Founding Intern | Xorvis AI Oct 2025 – Dec 2026

AI Agents for Chip Design and Verification

- Built a **domain-specific data corpus and database** covering RTL, verification, and chip-level workflows.
- Designed and implemented **agent systems** that transform design specifications into knowledge graphs, enabling structured reasoning over modules, signals, and verification intent.
- Developed **automated verification agents** that compile RTL, execute testbenches, and analyze simulation results by orchestrating EDA tools end-to-end.

Projects

Neural Solver for Dynamic Programming Problems (0/1 Knapsack) Nov 2025

Course Project – Deep Learning

- Built a **sequence-to-sequence neural model (BiLSTM with attention)** to learn dynamic programming table construction and solution decoding without explicit knowledge of problem constraints.
- Designed a **multi-task learning framework** to jointly predict DP tables, optimal values, and item selections, achieving up to **87.6% F1 score** on held-out problem instances.
- Developed **data augmentation strategies and adversarial test suites** to evaluate generalization, identifying key failure modes of neural algorithmic reasoning under distribution shift.

Advanced Functional Brain Imaging Analysis Tool Development Feb 2024 – Apr 2025

Prof. Rahul Garg, Dept. of CSE, IITD

- Developed comprehensive **Python command-line tools** for General Linear Model (GLM) analysis of fMRI data, including both single-subject and group-level statistical analysis.
- Implemented **neuroimaging preprocessing pipelines** including motion correction, spatial smoothing, temporal filtering, and registration to standard brain templates.
- Validated analysis results against **FSL (FMRIB Software Library)** outputs with correlation analysis and statistical comparisons across multiple subjects.

Personalized Offer Ranking | Amex Campus Challenge 2025 May 2025 - July 2025

American Express (Data Science Competition)

- Built a offer ranking model to predict $P(\text{click} \mid \text{impression})$ using customer, transaction, and offer metadata.
- Achieved a best **MAP@7 score of 0.567**, ranking the team in the **top 30** on the public leaderboard among nationwide participants.
- Optimized ranking performance under strict evaluation constraints, aligning model outputs with real-world engagement objectives.

Technical Skills

Programming Languages: Python, Java, C++, SQL, LaTeX
AL/ML: LLM Fine-tuning (SFT, DPO), Prompt Engineering, Reinforcement Learning, Transformers, Deep Learning
Tools/Frameworks: PyTorch, TensorFlow, LangChain, Hugging Face, NumPy, Pandas, Streamlit

Positions of Responsibility

- **Student Mentor - BSW**, Mentored 4 freshers Aug 2025 – May 2026
- **BRCA Rep - Hindi Samiti**, Himadri, BHM June 2023 – May 2024
- **Team Head**, RDV'23, BRCA June 2022 – May 2023
- **Organisation Committee MUN**, IITD MUN - RendezvousX June 2023 – May 2024